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PROBLEMS IN PALAEOMAGNETIC SYNTHESIS ILLUSTRATED BY RESULTS FROM PERMO-TRIASSIC DOLERITES IN GUYANA

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Relatively consistent palaeomagnetic results have been obtained after alternating field demagnetization of samples from seven of eight dolerite dykes (ten sites) intruding the basement shield in Guyana. Results from five of these dykes (seven sites, 33 samples) give an average site-pole at 63°S, 138°W, k = 24.6, α_{95} = 12.4°. Despite K–Ar ages ranging from 262 to 675 m.y., the good grouping suggests that these dykes are generally contemporaneous, and they are judged to be Permo-Triassic. The remaining two dykes have distinct, apparently Precambrian, magnetizations.

Comparison of these Permo-Triassic data with other studies of reputedly similar age rocks from elsewhere in South America, show a significant scatter, the cause of which cannot be uniquely determined. This problem is commonly encountered in general palaeomagnetic synthesis, and to alleviate it, the analysis of *all* site-poles from a single continent for a substantial time segment is recommended.

A polar-wander swath should result, the dimensions and pattern of which are theoretically a function primarily of secular variation and continental drift. Such a plot of site-poles gives appropriate weight to all useful palaeomagnetic data; it can be readily updated as new results are added to the data reservoir.

1. Introduction

Snelling and McConnell (1969) reported a K–Ar age of 450 ± 25 m.y. for a fresh dolerite dyke cutting the granite of the Guiana Shield in the southern part of Guyana. Permo-Triassic radiometric ages have been reported for dolerite dykes in Guyana and Surinam (Snelling and McConnell, 1969; Veldkamp et al., 1971) and the validity of the older age may now be questioned (Dalrymple et al., 1975). Initially, however, it was to obtain palaeomagnetic data from radiometrically dated, lower Palaeozoic igneous rocks from the Guiana Shield that a sampling of this and other dykes in Guyana was undertaken in 1971.

2. Sampling and measurement

The sampling sites are shown in Fig. 1, and the detailed magnetic results are summarized in Table I.

Core samples, 2.5 cm in diameter, were collected with a portable gasoline-powered drill, and oriented both with magnetic compass and a form of sun compass (astrocompass, La Rochelle, 1961), whereby the true meridian is obtained in the field. In the low latitude of the sampling area, a sun compass is particularly prone to error in orientation, especially around noon, but usually, the sun and magnetic compass orientations were in reasonable agreement (±3°). In one case (site *8*) where consistent, but anomalously large differences between the magnetic and astrocompass readings were obtained, an error in orientation of the latter is suspected, and the magnetic compass bearings (corrected for local declination) were used.

The outcrops of the dykes often consist simply of a linear concentration of boulders; although some boulders are very large, one could not always be certain that they represented solid undisturbed bed rock. A representative sampling of the "best looking outcrop" was always made and except for site *1*, this pro-

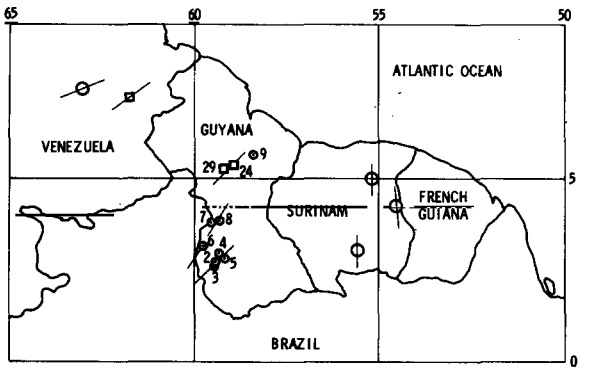


Fig. 1. Map of northeastern South America showing location of palaeomagnetic sampling sites and generalized trend of dolerite dykes at each site (short straight lines). Small circles numbered *2–9* are sites of samples collected in this study (Table I). Larger circles refer to the generalized sampling localities used in previous studies in Surinam (Veldkamp et al., 1971) and Venezuela (MacDonald and Opdyke, 1974); squares in Venezuela (El Callao) and in Guyana (numbered *24* and *29*) represent Permo-Triassic Minor Dyke Suite (Hawkes, 1966) localities sampled by Hargraves (1968). The apparent radial pattern of these dykes has been discussed by May (1971).

cedure has apparently given acceptable results. One unequivocally "in-situ" outcrop (site *2*) consisted of an extensive exposure of spheroidally weathered saprolitic dolerite in the Arakwai Creek. Cores were drilled in fresh in-situ residual spheroids at this exposure.

The natural remanent magnetism (NRM) was measured on a P.A.R. SM 1 spinner magnetometer with alternating field (AF) demagnetization on standard apparatus.

Selected cores from each site were systematically demagnetized in AF's up to at least 300 Oe, and in some cases up to 1000 Oe. Relative stability was established, and greatest consistency recorded after demagnetization in fields of 150, 200, or 300 Oe; all specimens (two from each core) were given this treatment at least. The site mean directions showed little difference between 200 and 300 Oe, but in each case, the best average, and the appropriate optimal demagnetizing field (ODF) is indicated in Table I.

Two of the four samples from site *3*, had anomalously strong NRM (0.2 e.m.u. cm^{-3}), which dropped dramatically on demagnetization. Their vectors converged on, but never reached the vectors from the other two samples, even in fields up to 1000 Oe, and therefore they are suspected of having been struck by lightning; so these two samples were excluded from the average of site *3*.

Sites *2* and *3*, and *4* and *5* are exposures of the

TABLE I

Summary of sampling site coordinates, reported rock ages, and magnetic properties measured

Name	Site No.	Coordinates (°'W)	(°'N)	*N*	*n*	*D**1 (°)	*I* (°)	*k*	α_{95} (°)	J_n (10^{-3} e.m.u. cm^{-3})
	1	59 22.5	2 50.5	4	4					
Arakwai	*2*	59 31.5	2 47.5	5	10	37.0	22.9	3.0	53.1	1.2
Arakwai	*3*	59 32	2 47.5	4/2	4	19.6	0.1	177.2	18.9	1.0
Kataliriwau	*4*	59 19.5	2 48.5	4	8	12.1	12.9	7.4	36.4	0.8
Kataliriwau	*5*	59 18	2 49	4	8	42.4	28.8	2.8	67.4	0.8
Mountain Point	*6*	59 39.5	2 59	4	8	14.2	−12.8	1.1	90	9.3
Moriru pond	*7*	59 32.5	3 54.5	5	10	359.4	−59.6	1.1	90	5.4
Puruwepi	*8*	59 19.5	3 55.5	5	10	93.4	−2.3	2.0	75.5	4.6
Hariwa Quarry	*9*	58 18.5	5 35.5	4	8	243.7	−21.1	13.0	26.5	4.6
Anarika	*10*	58 35	6 3	6	10	329.4	23.4	1.3	90	7.2

Table headings: Site number and coordinate; *N* = number of samples collected/used; *n* = number of specimens measured; *D* = declination; *I* = inclination; *k* = Fisher's precision parameter; α_{95} = half-angle of cone of 95% confidence; J_n = average intensity of NRM, 10^{-3} e.m.u. cm^{-3}; χ = average magnetic susceptibility, 10^{-3} e.m.u. Oe^{-1} cm^{-3}; MDF = median demagnetizing field, the demagnetizing AF required to reduce the NRM intensity by half; ODF = optimal demagnetizing field – that demagnetizing field at which the site mean remanent vector showed minimum dispersion. Apparent south poles are listed.

*1 *Note*: The first vector orientations and statistics given pertain to NRM; the second are results after AF demagnetisation.

same two individual dykes in different stream beds. While sites *2* and *3* are in good agreement, sites *4* and *5* are not. Possibly some of the very large blocks sampled at site *5* which were exposed on the bank of a stream, may have slumped to some extent, thus causing the difference. Nevertheless, for consistency, the results from both sites have been included in the average of Table I.

3. Petrography and magnetic properties

Petrographically, the dykes vary from fine- to medium-grained subophitic dolerites, with minor interstitial granophyre. The only exceptions to this are the dolerites at sites *6* and *9*. Site *6* contains some olivine and pigeonite inverted to orthopyroxene, in addition to augite and plagioclase. Site *9*, reputed to be a "Precambrian" dolerite (C.N. Barron, pers. commun., 1971), contains a distinctly pinkish pyroxene, with conspicuous biotite associated with opaques.

Observations in reflected light suggest that the predominant magnetic mineral in all dykes consists of titaniferous magnetite. These grains range in size up to 150 μm, and vary from roughly equant to skeletal. They invariably show fairly coarse exsolved ilmenite

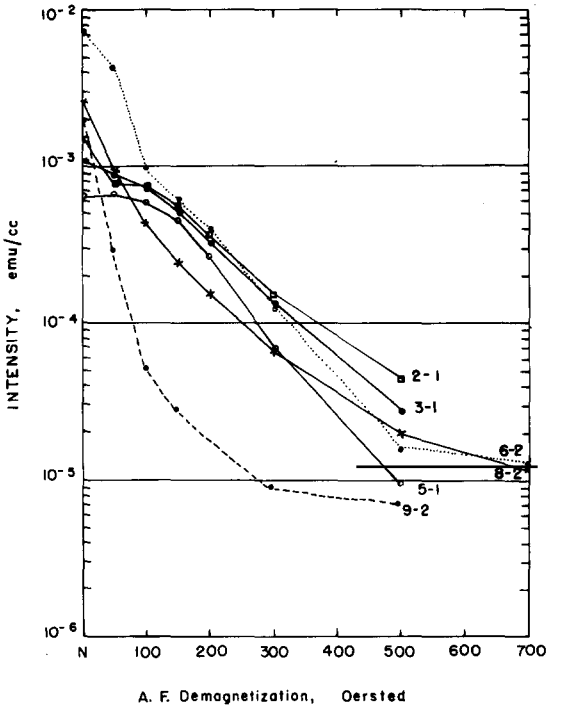


Fig. 2. Variation in intensity of remanent magnetism of Guyana dolerite samples with AF demagnetization. Numbe-- indicate site (as in Table I) and sample.

x (10^{-3} e.m.u. Oe^{-1} cm^{-3})	MDF (Oe)	ODF (Oe)	D*1 (°)	I (°)	k	α_{95} (°)	Pole lat. (°S)	Pole long. (°W)	K/Ar age *2 (m.y.)
3.4									
2.6	150	300	24.3	−12.6	563.8	3.2	64.1	128.7	450 ± 25 [1]
2.6	150	300	26.0	−2.4	275.7	15.1	63.7	141.1	450 ± 25 [1]
2.3	100/150	300	27.4	−12.0	217.8	6.2	61.2	131.3	262 ± 7 [2]
2.4	175	300	49.0	12.2	15.6	24.0	41.1	155.0	262 ± 7 [2]
2.3	50	500	42.4	−1.1	12.9	26.6	47.5	145.6	675 ± 22 [3]
2.7	50	500	312.9	−5.9	10.3	25.0	42.4	22.9 E	264 ± 11 [4]
1.4	50	300	7.5	−7.9	20.9	17.1	79.1	103.0	302 ± 8 [5]
1.9	25	200	287.1	−27.7	5.9	41.4	15.0	14.8 E	P_e
1.8	75	300	6.8	−1.8	23.4	14.1	80.3	103.1	undated

*2 *Source of age data*: [1] = Snelling and McConnell (1969); [2,3] = Guyana Geological Survey records, age data samples JPB 114 and JPB 107, respectively; [4,5] = Guyana Geological Survey records, C.N. Barron (pers. commun., 1975).

blades, and the residual magnetite is pervasively magnemitized.

The initial NRM of the dolerites is markedly reduced by AF demagnetization. The median demagnetizing field of the dolerites (MDF, listed in Table I) ranges up to 150 Oe, but for samples from several sites, it is 50 Oe, or less. Typical demagnetization profiles are shown in Fig. 2.

4. Age of the dolerite dykes

The available radiometric age data on the dykes sampled in this study are indicated in Table I. The K–Ar ages obtained from fresh dolerite dykes cutting crystalline basement rocks in Liberia (Dalrymple et al., 1975), were found to be variably *older* (in some cases considerably older) than ages obtained from petrologically and palaeomagnetically equivalent rocks cutting platform sediments. Dalrymple et al. (1975) consider the older ages to be spurious, and attribute them to excess Ar picked up from the crystalline country rock.

The K–Ar ages obtained from dolerite dykes in Guyana, Surinam and Venezuela, all of which cut Precambrian basement, should thus be treated as maximum ages and their temporal equivalence or otherwise might best be judged from the palaeomagnetic results. In the case of the Guyana dykes, in addition to those of apparent Permo-Triassic age, both the Arakwai dyke (sites *2* and *3*) dated at 450 m.y., and another from near Mountain Point (site *6*) dated at 675 m.y. all give closely similar magnetic poles (Table I). It is considered that these older K–Ar ages are also probably spurious, and that these dykes are all Permo-Triassic. Site *9*, the Hariwa quarry dyke, was thought beforehand to be Precambrian (C.N. Barron, pers. commun., 1971) and indeed, it carries a remanent vector analogous to that obtained from the Roraima dolerites (Hargraves, 1968; Veldkamp et al., 1971). The mean vector obtained from the site-7 samples, however, is much closer to other Precambrian vectors than to the Permo-Triassic direction. Both sites *7* and *9* have been excluded from the populations averaged in Table II.

5. Previous work

Previous palaeomagnetic studies of basic igneous intrusions in the Guiana Shield and platform have been reported by Hargraves (1968) and Veldkamp et al. (1971). The first of these studies was concerned primarily with Precambrian (ca. 1600 m.y.) intrusions into the Roraima sandstone, but included some data on samples from the reputedly Permo-Triassic "Minor Dyke Suite" (Hawkes, 1966). Veldkamp et al. (1971) reported on an extensive study of pigeonite–olivine dolerite dykes dated at 227 m.y. from eastern Surinam.

More recently, MacDonald and Opdyke (1974) combined results obtained from studies of a long dolerite dyke near Cerro Bolivar, Venezuela, dated at 199 m.y. (K–Ar), of rhyolitic and rhydacitic flows and tuffs of the Guacamayas Group at El Baul, Venezuela (195 m.y.) and of the Pescadero rhyolites from

TABLE II

Magnetic results obtained from volcanic tuff unit at base of La Quinta Formation, Venezuela (corrected for regional bedding dip: −55° in direction 290°)

Site	Coordinates		*N*	ODF (Oe)	*D* (°)	*I* (°)	*k*	α_{95} (°)	Mean pole *1* and *2* (*2* reversed)			
	(°'S)	(°'W)							lat. (°S)	long. (°W)	*k*	α_{95} (°)
1	8 9.5	72 01	5	150	6.7	−17.1	111.6	7.3	72	127	31.9	45.8
2	8 10.5	71 59	8	300	203.0	−7.3	103.6	5.5				
3	8 10.5	71 58.5	4	300	243	+19.1	300	5.3				

Abbreviations as for Table I.

Bedding attitude at site *3*, about 2 km W of a younger granite pluton, is most uncertain, and the regional value used is probably incorrect. Site *3* has been omitted from the calculation of the average pole.

Colombia, to give a mean Triassic pole for northern South America.

Hargraves and Shagam (1969) reported results from two samples of a volcanic tuff exposed at the base of the La Quinta Formation in Venezuela and concluded that this tuff was more likely to carry the original Triassic remanence than the La Quinta red beds with which they were principally concerned. Further sampling at the original and at two additional sites was subsequently undertaken (Shagam and Hargraves, 1973). No bedding correction could be made at one of the sites because the structural attitude was highly uncertain. Results from the other two sites are given in Table II.

Late Palaeozoic to early Mesozoic palaeomagnetic data derived from volcanic and sedimentary rocks mainly from Argentina and Brazil, have been discussed by Creer et al. (1969, 1970), Valencio and Vilas (1972), and Valencio et al. (1975). These last authors define a mean Triassic pole for South America (77°S, 121°W) which does not incorporate any of the published results from the Guiana Shield and platform, and they note that it is very close both to their Middle to Upper Permian pole, and to their Jurassic pole obtained from the Chon Aike Formation in Argentina. This led them to propose that South America was quasi-static (i.e. minimal apparent polar wander) from Middle Permian to Jurassic time.

6. Discussion

The pole positions obtained from studies of Permo-Triassic rocks from northern South America can be compared in Table III and Fig. 3 with the Triassic, and mean Permian to Jurassic poles of Valencio et al.

TABLE III

Summary of South American Permo-Triassic poles

Age		N	Lat. (°S)	Long. (°W)	k	α_{95} (°)
(a) Guiana Shield:						
Permo-Triassic	(*1*) Guyana dykes (this study)	7	63	138	24.6	12.4
Permo-Triassic	(*2*) Minor Dyke Suite (Hargraves, 1968)	3	71	178	80.8	13.8
227 m.y.	(*3*) Surinam dykes (Veldkamp et al., 1971)	10	82	40	25	10
199 m.y.	(*4*) Bolivar dykes, Venezuela (McDonald and Opdyke, 1974)	5	67	114	245.6	4.9
Mean (*1*) – (*4*)			74	133	27.3	17.9
(b) Other northern South American sites:						
195 m.y.	(*5*) Guacamayas Group, Venezuela (McDonald and Opdyke, 1974)	5	71	59	16.8	19.2
193 m.y.	(*6*) Pescadero Rhyolite Colombia (McDonald and Opdyke, 1974)	1	88	83	22.9	10.9
Triassic	(*7*) La Quinta Tuffs (Shagram and Hargraves, 1973; this study)	2	72	127		
Mean (*5*)–(*7*)			79.0	91.4	29.8	19.8
Mean (*1*)–(*7*)			77.2	118.9	32.0	10.8
(c) Southern South America (mainly Valencio and Vilas, 1972; Valencio et al., 1975):						
204 m.y.	SA Tr 1 Las Cabras Formation (Argentina)	5	74	94		(11,18)
224 m.y.	SA Tr 2 Ischigualasto–Ischichuca (Argentina)	9	79	121	13	15
232 m.y.	SA Tr 3 Puesto Viejo Formation (Argentina)	6	76	124	14	18
Triassic	SA Tr 4 Giron Formation (Colombia)	?	77	151		>16
Permo-Triassic	SA Tr 5 Paganzo Group (Argentina)	?	77	185		6
Mean of SA Tr 1–SA Tr 4 (Valencio et al., 1975)		(4)	77	121	190	66
Mean of Middle Permian to Middle Jurassic poles (Valencio et al., 1975)		(9)	81	128	131	4

Minor Dyke Suite pole is average of three sites labelled (Fig. 1) Nos. *9* (Laguna Mine, El Callao, Venezuela), *24* (Tumatumari, Guyana) and *29* (Potaro River bridge, in Guyana)

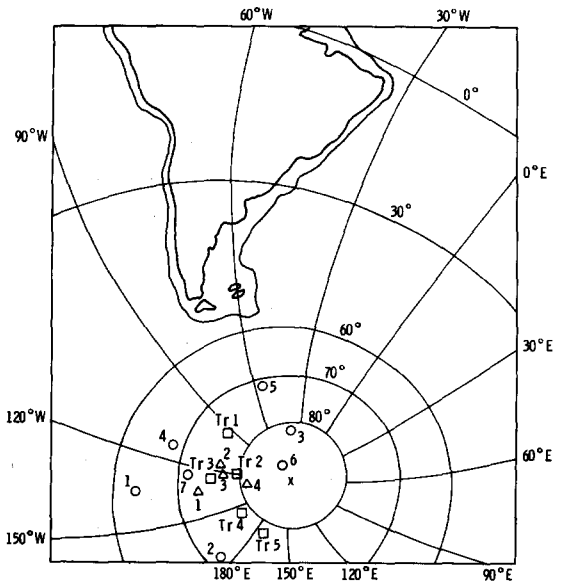


Fig. 3. South American Permo-Triassic poles and mean poles, as in Table III. Circles from northern South America; squares from southern South America (mainly); triangles, are mean poles: *1* = mean of Guyana dolerites ($N = 4$) alone; *2* = mean for northern South America ($N = 7$); *3* = mean Triassic pole; and *4* = Middle Permian to Middle Jurassic poles; *3* and *4* are from Valencio et al. (1975).

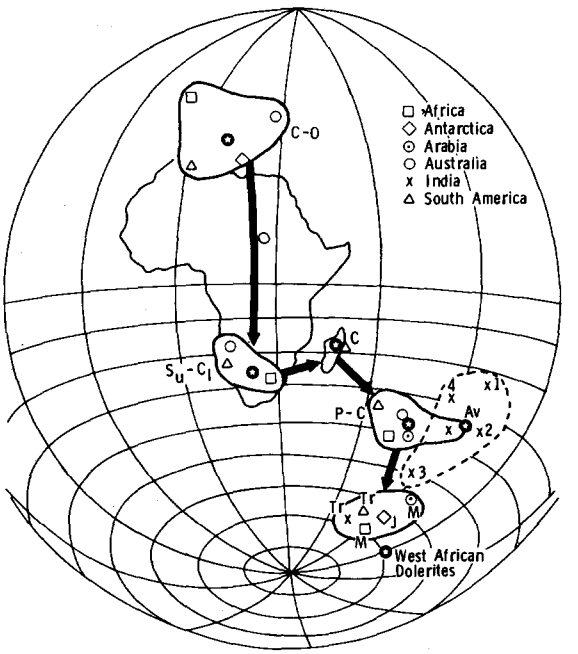


Fig. 4. Pole-groups and generalized polar-wander curve for Gondwana, Cambrian to early Mesozoic (from McElhinny and Embleton, 1974). A broken line encloses the group of Guiana Shield dolerite poles and their mean (crosses numbered *1–4*, *Av*; Table III), after rotating South America back to Africa according to the model of Bullard et al. (1965). The West African dolerite pole is that of Dalrymple et al. (1975) dated at 190 m.y. On the Gondwana curve: *C–O* = Cambro-Ordovician; S_u–C_l = Upper Silurian–Lower Carboniferous; *P–C* = Permian-Carboniferous; *M*, *Tr* and *J* = Mesozoic, Triassic and Jurassic; open star in solid circle signifies group average.

(1975), from studies in Argentina and Brazil. Despite the scatter of the Guiana Shield and platform results, the mean is very close to that of Valencio et al. (1975). This could be regarded as confirmation of the quasi-static condition of South America inferred by them for the interval Permian–Jurassic. However, as is also generally true for the individual poles used in the averaging of Valencio et al. (1975), the Guiana Shield poles listed and averaged in Table III are each purported to be valid palaeomagnetic poles, not virtual geomagnetic poles. The scatter of these poles nominally indicates that the Guiana Shield at least (if not the whole continent) was drifting, or that the pole was wandering, during that interval. This problem of how best to treat scattered, but nominally contemporaneous, palaeomagnetic poles from what is assumed to have been one rigid continent is considered in Section 7.

After rotation of South America back to its predrift position relative to Africa (Bullard et al., 1965), the Guiana Shield dolerite poles and their mean are compared in Fig. 4 with the relevant section of the average polar-wander curve for Gondwana of McElhinny and Embleton (1974). While the individual poles are of course scattered, their mean falls near, but not on, the Permo-Carboniferous (*P–C*) elbow in the Gondwana curve. Also shown in Fig. 4 is the well dated (190 m.y.) pole of Dalrymple et al. (1975), obtained from numerous dolerite dykes in West Africa. This pole is quite distinct from all of the Guyana dolerite poles which would therefore appear to be older than 190 m.y. On the other hand, all the Guyana dolerites are normally magnetized, suggesting that they are younger than the predominantly reversed Kiaman interval, which ended about 235 m.y. ago (Thompson and Mitchell, 1972). If this is correct, then taking the angular separation of the poles as 30°, and the maxi-

mum age difference of 45 m.y., we obtain an average drift rate of over 6 cm year^{-1}.

The West African dolerite pole of Dalrymple et al. (1975) is not contained within the Mesozoic group of Gondwana poles compiled by McElhinny and Embleton (1975) (Fig. 4). Again it is not clear how best to regard this difference, and I consider that this fundamental issue is evaded by simply incorporating this result in the Mesozoic average.

7. Synthesis of palaeomagnetic data

The scatter in the purportedly contemporaneous poles listed in table III poses a frustrating problem common to synthesis of palaeomagnetic studies. Vector means of all or of parts of these data are given in Table III. The overall mean pole, however, is not necessarily the best estimate of the Permo-Triassic palaeopole for South America.

Some considerations which must be borne in mind when any judgment of the acceptability of palaeomagnetic data is made, are as follows.

(1) It is possible that systematic experimental errors were introduced in field or laboratory orientation. All the results reported are from samples which have been subjected to magnetic washing, and without interlaboratory comparison of measurements made on the same sample, one cannot "a priori" select measurements made in one laboratory over another: such errors may be present, but the long-term refinement of palaeomagnetic data cannot depend upon a subjective suspicion of such errors. Hence *all* the data must be used.

(2) In palaeomagnetic studies of plutonic rocks, or of shallow intrusives into crystalline basement, the original attitude at the time of cooling cannot be uniquely established. There may be systematic errors in individual palaeomagnetic studies due to such indetectable tectonic causes. But as long as they remain undetectable, such erroneous data will continue to become incorporated in the record, unless one chooses to exclude data from such environments.

(3) There are large differences in the number of samples/sites incorporated in the various studies listed in Table III. Other things being equal, a pole based on a study of 100 samples from 20 sites, should be superior to one based on a study of 20 samples from five sites, and any mean of the two poles should be weighted in favour of the more detailed study, taking note of the number of sampling sites involved.

(4) Viewed in the most optimistic light, the scatter of data in Table III could be attributed to differences in the time of magnetization. On the one hand, secular variation may not have been averaged out, and on the other, the magnetizations may be different in age by millions of years or more. During such a considerable interval, apparent polar wandering or continental drift may in fact have taken place. Granted the extreme difficulties attending the unambiguous determination of *when* magnetization was acquired, (even with radiometric age determinations) and uncertainty as to the duration of the magnetizing interval represented, it is almost impossible to determine objectively, whether a difference in pole position is due to secular variation or continental drift!

In light of the above uncertainties, a best mean pole may be obtained after minimum stability criteria are satisfied, by judicious, semi-subjective selection of data. An alternative, more laborious approach, which involves the *analysis of all penecontemporaneous palaeomagnetic data at the site-pole level*, is advocated here.

7.1. Site-pole patterns

If results from several samples at a single "site" give reasonably consistent results, then I assume that this represents a palaeomagnetic vector imposed by the earth's magnetic field.

At its *worst*, from a palaeomagnetic point of view, the site vector represents the earth's field only at an instant of time. A collection of all such site vectors over a significant interval of time should display a circular distribution if no drift occurred, or an elliptical (or more complicated) distribution pattern if drift did occur. In either case, the angular diameter of the circle, or the minor axis of the ellipse should be a *function of magnitude of secular variation*. The major axis of an elliptical distribution should in addition be *a function of the amount and direction of apparent polar wander*. In such a manner, the practice of giving unit weight to all site poles allows the incorporation of all penecontemporaneous palaeomagnetic data into one single population for one continent. The fact that the distribution need not be circular allows more un-

certainty in the absolute dating and duration of the time of magnetization.

Ideally, the incorporation of all site poles, for a continent, for the entire Phanerozoic for example, should define a polar-wander swath for that continent. Loops in the polar-wander path could obviously be obscured by this treatment, but proof of such a loop requires accurately-dated palaeomagnetic data with small scatter; the subjective selection of data deemed to meet these requirements is not precluded by the first-order synthesis of *all* site-poles advocated here. A synthesis and comparison along these lines of all published Permo-Triassic palaeomagnetic data from South America, Africa and North America, are in progress.

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